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Federal Wage System Series

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CAREER FIELD EDUCATION AND TRAINING PLAN

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Part I

Preface

This Career Field Education and Training Plan (CFETP) is a comprehensive education and training document that identifies life-cycle education/training requirements and training support resources for the identified series. The CFETP provides Federal Wage System (FWS) personnel with a clear career path to success and instill rigor in all aspects of our job series training.

The CFETP consists of two parts used by the supervisor to plan, manage, and control training within the job series.

Part I includes the following:

- **Section A** provides general information about how the CFETP will be used.
- **Section B** identifies job series progression information, duties and responsibilities, training strategies, and the job series path.

Part II includes the following:

- **Section A** identifies the Family Series Training Standard (FSTS) to include duties, tasks, and technical references to support FWS training programs.
- **Section B** identifies available support materials.
- **Section C** identifies a training course index supervisors can use to determine resources available to support training. Included here are both mandatory and optional courses, and exportable courseware.

Using guidance provided in the CFETP will ensure individuals in the identified series receive effective and efficient training at the appropriate point in their careers. This plan will enable us to train today's workforce for tomorrow's jobs. At the unit level, supervisors and trainers must use Part II to identify, plan, and conduct training commensurate with the overall goals of this guide and the local mission.

Section A: General Information

A1. Purpose

The CFETP is designed to be a tool for supervisors to use in assessing the skill level of current and new employees. The CFETP may be used to document training and proficiency of the employee on associated task/s by the supervisor or certified trainer. The FSTS (Attachment 2) portion of the CFETP is designed to be a tool for supervisors to use in assessing the skill level of current and new employees. The CFETP may be used to document training and proficiency of the employee on associated task(s) by the supervisor or certified trainer.

A2. Uses

Managers and supervisors may use the plan at all levels to ensure comprehensive and cohesive training programs are available for everyone in the FWS series.

A2.1. Federal Wage System Panel of the Functional Advisory Council

The Federal Wage System Panel of the Functional Advisory Council will develop/revise formal resident, non-resident, field, and exportable training based on requirements established by the users and documented in Part II of the CFETP. They will also work with the Air Force Civil Engineer Center Force Development Division (AFCEC/COF) to develop acquisition strategies for obtaining resources needed to provide the identified training.

A2.2. Federal Wage System Panel along with the Readiness and Training Managers

The Federal Wage System Panel along with the Readiness and Training Managers, will ensure their training programs complement the CFETP training requirements and identify requirements that can be satisfied by OJT, resident training, contract training, or exportable courses.

A2.3. Supervisors

Supervisors will guide individuals through completion of training specified in this plan.

A2.4. Individuals

Individuals complete training requirements specified in this plan. The list of courses in Part II of this CFETP will be used as a reference to support training.

A3. Coordination and Approval

The Air Force Civil Engineer Civilian Career Field Manager is the approval authority for the CFETP. The Federal Wage System Panel along with the Readiness and Training Managers will identify and coordinate on FWS series training requirements. Using the list of courses in Part II, they will eliminate duplicate training.

A4. Head Quarters United States Air Force (HQ USAF)/A4C

HQ USAF/A4C will review this CFETP annually and make updates and changes as deemed appropriate. Please send recommended changes to the AFCEC/COF Training Support Section at DSN 523-6879 or comm. 850-283-6879 or email.

Section B: FWS Progression and Information

B1. Series Descriptions

See specific Standard Core Personnel Document (SCPD) for the description.

B1.1. Specialty Summary

Constructs and maintains concrete and asphalt runways, aircraft parking aprons, and roads. Operates and maintains heavy construction equipment, such as loaders, graders, dozers, backhoes, and dump trucks. Operates tractor-trailer combinations, transporting construction equipment, and materials. Ensures compliance with all safety and environmental regulations. Related DoD Occupational Subgroup: 171000.

B1.2 Commercial Driver's License Requirements

Establishing Commercial Driver's License (CDL) requirements for select positions within the Pavements and Construction Equipment Operator Wage Grade specialty ensures personnel are properly qualified to operate mission-essential vehicles safely and in compliance with federal and Air Force directives. This requirement supports operational effectiveness, risk reduction, and workforce readiness. At the unit level, supervisors and trainers must review and understand applicable Air Force guidance to accurately identify positions requiring a CDL, and to plan, implement, and sustain training and licensing in alignment with regulatory requirements and mission needs.

B2. Skill and Career Progression

It is essential everyone involved in training do their part to participate in, plan, manage, and conduct effective training. The guidance provided in this part of the CFETP will identify viable training at appropriate points in an individual's career.

B2.1. Apprentice/Helper (A/H)

B2.1.1. Upon completion of initial skills training, an employee may work with a trainer to enhance their knowledge and skills to perform at the highest attainable level within their series.

B2.1.2. Utilize formal training, the CDC and other exportable courses for subject and task fundamentals in the series.

B2.1.3. Encourage apprentice/helpers to continue academic education and begin Enlisted Professional Military Education (EPME) by enrolling in Airman Leadership School either in-residence or by correspondence course.

B2.2. Journeyman (J)

B2.2.1. Journeymen may enhance their skills by engaging in additional training. After completing training, they may be assigned job positions such as team leader, trainer, or task certifier. Journeymen can pursue leadership training and skills to qualify for potential advancement to Work Leader or Work Supervisor positions.

B2.2.2. Encourage journeyman to enroll in the Noncommissioned Officer Academy (NCOA) either in-residence or by correspondence course.

B2.3. Craftsman (C)

B2.3.1. Craftsmen may continue to advance their skills by completing additional training. They may be assigned job positions such as team leader, trainer, or task certifier. Craftsmen are encouraged to pursue leadership training and skills to qualify for potential advancement to Work Leader or Work Supervisor positions.

B2.3.2. Encourage craftsmen to continue academic education and complete NCOA either in-residence or by correspondence course, civilian leadership courses and degree programs.

B2.3.3. Master Craftsman are typically graded higher than WG-10 where skills, knowledge and abilities require higher technical abilities than standard craftsmen. They are duty/location specific and not for all job series.

B2.4. Work Leader (WL)

B2.4.1. Work Leaders are expected to perform limited supervisory duties or act as a team lead.

B2.4.2. Completion of the Air Force Institute of Technology (AFIT) Civilian Supervisors Course (WMGT 571) is highly encouraged.

B2.4.3. Should pursue increased knowledge of budget, manpower, resources, and personnel management.

B2.4.4. Recommend pursuit of additional higher education and completion of courses outside of their job series for career broadening opportunities.

B2.4.5. Encourage Work Leaders to continue academic education and complete NCOA either in-residence or by correspondence course, civilian leadership courses and degree programs.

B2.5. Work Supervisor

B2.5.1. A supervisor can be expected to fill positions such as the Element Chief or Special Projects Supervisor.

B2.5.2. Must enroll and complete required mandatory supervisor or manager training courses within 1 year of appointment to a supervisory or managerial position and complete experience training every 3 years, thereafter.

B2.5.3. Completion of the AFIT Civilian Supervisors Course (WMGT 571) is highly encouraged.

B2.5.4. Should pursue increased knowledge of budget, manpower, resources, and personnel management.

B2.5.5. Recommend pursuit of additional higher education and completion of courses outside of their job series for career broadening opportunities.

B2.5.6. Encourage supervisors to continue academic education and complete Senior Noncommissioned Officer Academy (SNCOA) by correspondence, Emerging Leader course and degree programs.

B3. Correspondence Course Directions

Nonresident attendance for professional military education courses is accomplished through the Air Force (AF) Portal.

B3.1. Step 1

Login to the AF Portal.

B3.2. Step 2

Copy and paste the Air University URL into your browser.

B3.3. Step 3

Choose the "Distance Learning" tab on the on the right-side menu.

B3.4. Step 4

Select the appropriate course.

B4. Enlisted Professional Military Education (EPME)

Scheduling enlisted professional military education for civilian personnel is a responsibility of AFPC/DP3DW unless otherwise noted. See Department of Air Force Instruction (DAFI) 36-2670, *Total Force Development*, for more detailed information.

B4.1. Airman Leadership School (ALS)

Airman Leadership School resident attendance is scheduled by the local ALS commandant. Commandants build an annual schedule shortly after the staff sergeant (E-5) promotion release. Eligible AF and Department of Defense (DoD) civilians are considered priority 3 for ALS. DAFI 36-2670 encourages ALS leadership to the extent possible, distribute Priority 2 and 3 students throughout the year to maximize diversity in the classroom.

B4.2. Noncommissioned Officer Academy (NCOA)

Selection and scheduling are accomplished by AFPC/DP3DW. Eligible AF and DoD civilians are considered priority 3 for NCOA and are considered on a space available basis.

B4.3. Senior Noncommissioned Officer Academy (SNCOA)

Selection and scheduling are accomplished by AFPC/DP3DW and is driven by the senior master sergeant (E-8) promotion release. Eligible AF and DoD civilians are considered priority 4 for SNCOA and are considered on a space available basis.

B5. Career Building Blocks (CBBs) and Continuous Development Framework Model

B5.1. Career Building Blocks (CBBs)

The CBBs illustrate the dispersion of grades and relative experience levels (apprentice, journeyman, craftsman). As you progress through the grades, so should your breadth of experience. This not only includes technical experience, but leadership, supervisory and management experience as well. Using various on-the-job training, military, and civilian vocational schools, and professional military education (PME), forms the foundation of the CBBs and the "Develop Exceptional Leaders" portion of the strategic vision in the Civil Engineer (CE) Human Capital Road Map.

B5.1.1. CBBs

The CBBs are an illustration or an example of your individual development plan or career path. Your individual roadmap may be different than others based on your occupational series and/or career path. The CBBs provide a few examples of development opportunities at the tactical, operational, and strategic levels of performance. Click here to view the CBBs for your job series.

B5.1.2. Career Goals

Once you have established your career goals, identify a mentor to help you align your steps in this learning and development continuum. A mentor can help you identify different kinds of experience and training you will need as well as the tools to attain them. Mentors also serve as a sounding board and can help you reassess or adjust your career goals when events in your life or career necessitate changes. As you complete these steps and move toward your career goals, it is important to reassess your goals. Once this is accomplished, you and your mentor will be able to further your progression and advancement through the continuous development framework model. Refer to Figure 1 (below) for visual reference.

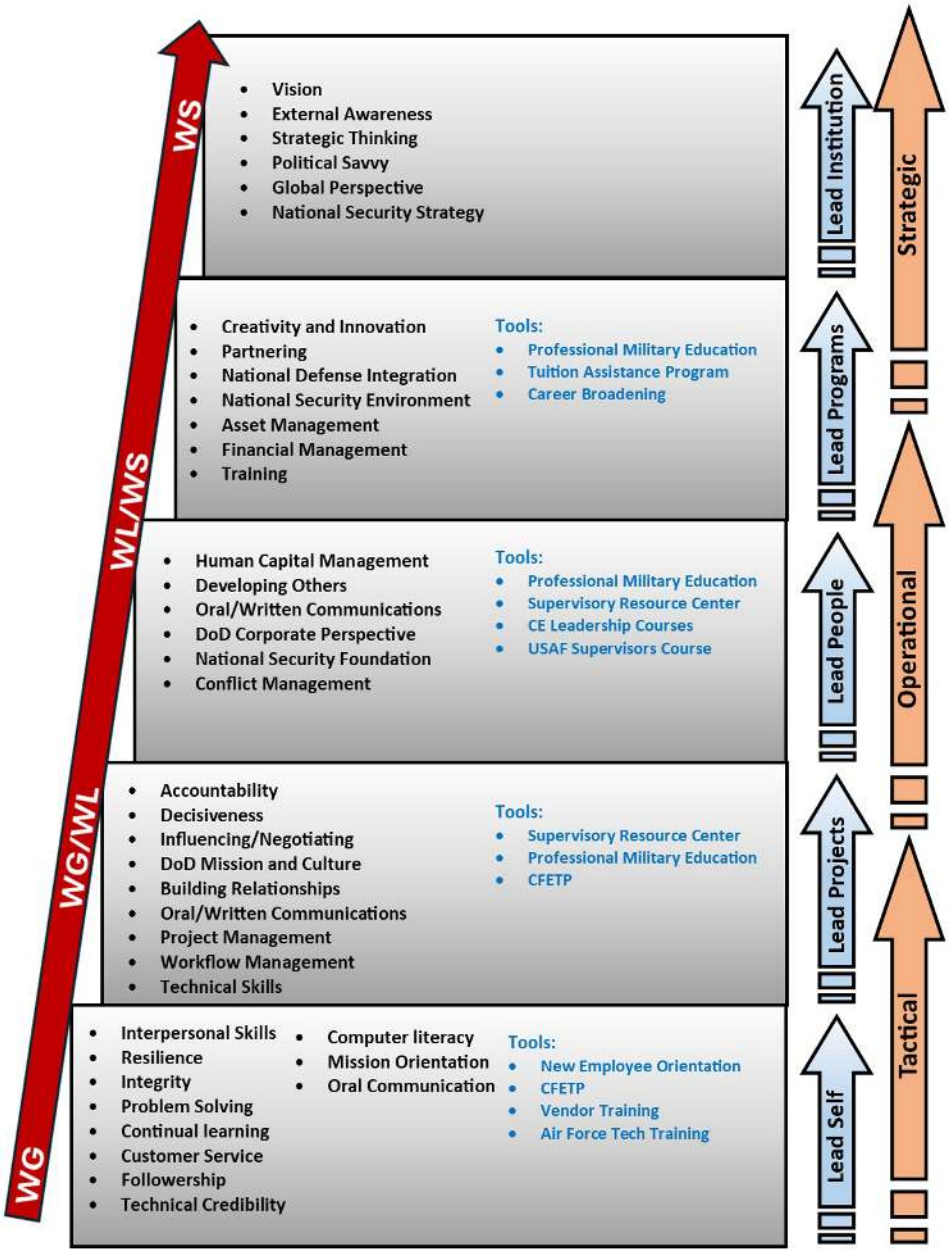


Figure 1: Continuous Development Framework Model

Part II

Section A: Family Series Training Standard (FSTS)

A1. Implementation

This FSTS is used for technical training provided by Air Education and Training Command (AETC), OJT and COTS training documentation.

A2. Purpose

As prescribed in DAFI 36-2670, *Total Force Development*, DAFMAN 32-2670, *Civil Engineer Federal Wage System Force Development* and in collaboration with the Air Force Civilian Career Field Manager (AFCCFM), this FSTS is mandatory for all FWS job series working under the AF Specialty Code 3E2X1, regardless of duty assignment. Each employee must use an automated training record.

A2.1. Column 1 (*Tasks, Knowledge, and Technical References*)

Lists the most common performance and knowledge requirements and supporting technical references (TR) necessary for an employee to perform duties in the Apprentice, Journeyman, Craftsman, and Supervisor level successfully. The required behaviors will be used as the context for which learning will be assessed.

A2.2. Column 2 (Core Tasks)

The CFM or shop foreman is responsible for identifying the mandatory tasks for each duty position within a job series at their specific location. Tasks identified have Core and/or Certification requirements, Civilian Deployment needs, or those designated with a Special Experience Identifier (SEI).

A2.2.1 Column 2 (*Deployment/SEI*)

Tasks identified as mandatory for employees who perform roles in Mission Critical, Mission Essential or have been identified for a civilian deployment tasking.

A2.3. Column 3 (Provides Certification for OJT)

Used to record completion of tasks and knowledge training requirements. Task certification requires the task to be trained by a trainer designated by the supervisor. The trainer can be either civilian or military. The training start and completion date are documented, the task is signed by the employee and either the work center supervisor, a Staff Sergeant (or above), Civilian equivalent, or the unit training manager. This action will complete the task certification.

Note: The "trainer" signing the record MUST be the work center supervisor, work leader, a Staff Sergeant (or above), Civilian equivalent, or the Unit Training Manager. This person does not necessarily train the task but will ensure the training is conducted by a qualified trainer prior to completing task certification.

Note: If a work center supervisor, Work Leader, a Staff Sergeant (or above), Civilian equivalent, or the unit training manager are not available in a shop or unit to certify a task, the Operations Flight deputy commander will designate a certifier within the flight.

A2.3.1. Task Certification

Task certification of core and critical tasks. Require a training completion date and initials of the employee, trainer, and a certifier. All non-core tasks require training completion date and initials of the employee and trainer only.

A2.3.2. Performance Standard

All training requirements are trained and qualified to the "Go" level. "Go" means the individual can perform the task without assistance and meets local demands for accuracy, timeliness and, if applicable, correct use of procedures and Technical Orders.

A2.4. Column 4 (Codes Used to Indicate level of Training)

Indicates whether the task is a Knowledge (K), Performance (P) or Performance and Knowledge (PK). Codes are provided in columns labeled for each level of experience. These are labeled A/H for Apprentice/Helper, J for Journeyman, C/WL for Craftsman or Work Leader and S for Supervisor. Identifies duty position tasks (series training requirements) with a proficiency code and indicates training requirements. It shows the proficiency to be demonstrated on the job by the employee as a result of hands-on training on the task, knowledge and the career knowledge provided by formal courses, career development courses (CDCs), distance learning (DL) web-based training (WBT) and Air Force Qualification Training Packages (AFQTPs). CDC listings are maintained by the unit training manager (UTM) for current CDC listings.

A2.5. Qualitative Requirements.

Contains the proficiency code key used to indicate the level of training and knowledge provided by computer-based training (CBT), WBT, commercial off the shelf (COTS), in-resident training, and career development courses.

A2.6. Job Qualification Standard (*JQS*)

The FSTS becomes the Job Qualification Standard (JQS) for on-the-job training (OJT) when entries are made in the FSTS. For OJT, the tasks in Column 1 are trained and qualified to the go/no go level. "Go" means the individual can perform the task without assistance and meets local requirements for accuracy, timeliness, and correct use. JQS is used according to DAFI 36-2670, *Total Force Development* and DAFMAN 32-2670, *Civil Engineer Federal Wage System Force Development*. AFQTPs are available on myLearning to ensure that all supervisors use standardized procedures for training. When used as a JQS, the following requirements listed below apply.

A2.6.1. Documentation

Document and certify completion of training.

A2.6.1.1. Duty position

Requirements for each duty position (task group) will be developed and identified by the work center supervisor and loaded into the automated training

management application. Completion of the identified tasks is mandatory for all duty positions. Ensure the correct duty position title is listed in the Profile section of the employee's automated training record.

A2.6.1.2. AFQTP Training and Documentation

AFQTPs have been created for several task groups to fulfill performance (P) and knowledge (K) requirements for upgrade/qualification training. Each AFQTP provides the step–by–step procedures for the trainee, trainer, and certifier in completing each.

A2.6.1.2.2. Hands-On Training

For performance (P) training requirements, DO NOT sign off the tasks in the JQS until the employee has completed hands-on/ certification training.

A2.6.2. Transcribing from previous versions to new CFETP

The UTM and supervisor must conduct a review of the new STS to identify any new tasks and add those tasks to their unit specific duty positions.

A2.6.2.1. Previous training certification not listed

If previous training certifications are not listed in the individual training record, select the task to be transcribed, and click on the transcribe button. Enter the date of the original certification and sign off the task(s). The employee will then sign off the task(s) to finalize the transcription of previous training certification. The automated application will place an entry into the employee's training record and must be acknowledged by the transcriber and employee.

A2.6.2.2. Transcribing external training certification

If an employee attended a formal training course and received appropriate accreditation, select the formal training section of the users automated training record and locate the course title in the master task list, then enter the completion date. If the course is not listed, contact the Unit Education Training Manager (UETM) to have it loaded from the master catalog. If it is not listed in the master catalog, contact AFCEC/COF to have it loaded in the master catalog.

Section B: Course Objective List

B1. Measurement

Measurement of each learning objective is indicated as follows:

B1.1. Use of Progress Checks (PCs) & Rubrics

Indicates formal measurement of knowledge (K) and/or performance (P) elements.

B2. Standard

Standards for measurement are indicated in the course objectives and delineated on the individual progress checklist and rubrics. The minimum standard is 70% on knowledge progress checks. Trainer assistance is used as the standard for performance progress checks and is provided, as warranted during the progress check. Employee may be required to repeat all or parts of the learning outcome until satisfactory performance is attained.

B3. Proficiency Level

Student must demonstrate mastery on each learning outcome/objective before progressing to the next learning requirement.

B4. Course Objective List

These objectives are listed in the sequence taught by Blocks of Instruction. Per Air Education and Training Command Instruction (AETCI) 36 - 2651, *Basic Military and Technical Training*, a detailed listing of the initial skills course learning objectives in the Basic Course are listed in the 3E4X1 Air Force Specialty Code (AFSC) STS.

Section C: Support Material

C1. Air Force Qualification Training Packages (AFQTP)

C1.1. AFQTP Documentation Record

The AFQTPs for each task group are identified on the AFQTP Documentation Record located in Attachment 3.

C1.2.1. Applicable AFQTPs

For a complete list of up-to-date applicable AFQTPs go to myLearning.

C1.2.2. Web-based Courses

In addition to the AFQTPs, there are web-based courses or assessments developed for certain tasks that are available on myLearning under AFCEC in the specialty topic area.

C2. Career Development Course (CDC) Assessment for CE CDC/DL Course

C2.1. CDC Assessments

AFCEC/COF has developed CDC assessments for each AF CE career field, and they are located on the myLearning under AFCEC in the topic header CE CDCs Assessments.

C2.2. CDC Purpose

CDC assessments are for the sole purpose of providing the trainer and the supervisor a predictive indicator of whether the employee has a solid grasp on the knowledge portions of the STS.

Section D: Education and Training Course Index

D1. Purpose

This section of the CFETP identifies training courses available for the [**Insert specialty name**] Specialty. Refer to Education and Training Course Announcements for information on the AF in-residence courses.

D2. Air Force In-Residence Courses/Mobile Training Team (MTT) Courses

Course Number	Title	Developer
JCAQP3E231 00AB	Pavements and Construction Equipment Apprentice	368TRS
J9AQA3E231 04RA	Pavements and Construction Equipment Operator (ITRO Graders)	368TRS
JCABP3E231 00AE	Pavements and Construction Equipment Operator Apprentice	368TRS
JCACP3E271 00AA	Pavements and Equipment Craftsman Course	368TRS
J5AZA3E251 00RA	Sling Load Inspector Certification (860-F4)	368TRS
AFCEC 3E2X1	Regional Equipment Operator Training School	201 RTS

Table 1: AF In-Residence Courses/MTT Courses

D3. Air Force Career Development Academy (AFCDA) Courses

Course Number	Title	Edit Code
CDC 3E251	Pavements and Construction Equipment Operator Journeyman	02

Table 2: AFCDA Courses

D4. Exportable/Web-based Courses/Information

Course Number	Title	Developer
Web based	Civil Engineer 5-Level Core Concepts Course	AFCEC/COF
Web based	Civil Engineer 7-Level Core Concepts Course	AFCEC/COF
Web based	Airfield Crushed Stone Crater Repair	AFCEC/COF
Web based	Airfield Rubber Removal	AFCEC/COF
Web based	Paved Surfaces (Rigid & Flexible)	AFCEC/COF
Web based	Snow and Ice Control Techniques	AFCEC/COF
Web based	Snow and Ice Control Management	AFCEC/COF
Web based	Spall and Crack Repair	AFCEC/COF
Web based	Dump Truck Operator's Course	AFCEC/COF
Web based	Fall Protection Awareness	AFCEC/COF
Web based	Tractor Trailer Combination	AFCEC/COF
Web based	Tractor Trailer Overview	AFCEC/COF
Web based	Air Brake Testing	AFCEC/COF
Web based	Oshkosh H Series Cat C 13 Regeneration Procedures	AFCEC/COF
WMGT 322	Introduction to Project Management	AFIT
WMGT 301	Intro to Asset Management	AFIT
WENG 200	Scoping and Estimating	AFIT
WENG 436	Requirements and Optimization	AFIT
WENG 437	Troop Construction Project Management Course	AFIT

Table 3: Exportable/Web-based Courses

D5. Leadership Training and Academic Fellowships through the Eaker Center

Title:

- Squadron Officers School (SOS)
- Developing Supervisor Course
- Defense Emerging Leader Program
- Civilian Associate's Degree Program
- Civilian Bachelor's Degree Program
- Civilian Master's Degree Program

Located at: Civilian Force Development Homepage

Attachment 1 – Qualitative Training Requirements (Proficiency Code Key)

A1. Qualitative Training Requirements

This Block is for Identification Purposes Only.		
Name Of Employee		
Printed Name (*Last, First, Middle Initial*)	**Initials (*Written*)**	**SSAN (*Last four*)**
Printed Name of Trainer, Certifying Official and Written Initials		
N/I	*N/I*	

Attachment 2 – FWS Family Series Training Standard (FSTS)

A2. Family Series Training Standard (FSTS)

A2.1. Identification

In the training record User Profile section, the UTM will assign individuals to the correct work center, upon in-processing into the unit.

A2.2. Specialty Tasks

The following are tasks the work center supervisor will use to track each duty position created for their work center.

Behavioral Statement FSTS Coding System	
Code	**Definition**
K	Subject Knowledge Training - The verb selection identifies the individual's ability to identify facts, state principles, analyze, or evaluate the subject.
P	Performance Training - Identifies that the individual has performed the task to the satisfaction of the trainer/certifier; however, the individual may not be capable of meeting the field requirements for speed and accuracy.
PK	Performance Knowledge Training - The verb selection identifies the individual's ability to relate advanced facts, procedures, operating principles, and operational theory for the task.
X	Formal Course
*	Core Task. These are mandatory tasks identified by the Career Field Manager

Table 4: Behavioral Statement FSTS Coding System

See the Federal building competency model

1. Tasks, Knowledge and Technical References	2. Tasks	3. Certification of OJT					4. Proficiency Codes			
	Core/Cert	A	B	C	D	E	A	B	C	D
		Tng Start	Tng Complete	Trainee Initials	Trainer Initials	Certifier Initals	Apprentice/ Helper	Journeyman	Craftsman Work Leader	Supervisor
1. CIVIL ENGINEER (CE) CORE CONCEPTS COURSE: TR: CE Virtual Learning Center (CE-VLC); AFDD 2-4-2; AFIs 10-209, 10-210, 10- 211, 32-1022, 36-2101, 38-101, 51-903; Enlisted Classification Directory; AF PAM 32-1004 Vol 1-6; War Mobilization Plan (WMP) 1, Annex S										
1.1. Civil Engineer (CE) 5-Level Core Concepts Course							K	K	K	K
1.1.2. Civil Engineer (CE) 7-Level Core Concepts Course									K	K
1.2 IN-RESIDENT AND DL CORE DUTY COURSES										
1.2.1. WMGT 141 SMS Paver Level 1								X	X	
1.2.2. WMGT 241 SMS Paver Level 2									X	X
1.2.3. WMGT 301 Intro to Asset Management							X	X		
1.2.4. WMGT 322 Intro to Project Management								X	X	
1.2.5. WENG 200 Scoping and estimating									X	X
1.2.6. WENG 436 Requirements and optimization								X	X	X
1.2.7. WENG 437 Troop Construction Project Management								X	X	
1.2.8. Civilian Supervisor Course										X
1.2.9. WMGT 571 Course									X	X
2. Contracting Officer Representative (COR) Duties: TR: Federal Acquisition Regulation (FAR) Part 37.6, 46; AF FAR Supplement Part 5346; AFI 63-138; AFPAM 32-1006										
2.1. Duties and Responsibilities									pk	pk
2.2. Knowledge of contract construction										
2.2.1. Project planning phase									K	K
2.2.2. Pre-project contractor walk through									K	K
2.2.3. Evaluate contractor's performance									K	K

3. AFS Specific Safety: **TR: AFIs 91-202/203/207, AFOSH STDs 48- 137, 48-20, 29 CFR 1910 subpart I**										
3.1. Use protective equipment										
3.1.1. Eye protectors							pk	pk	pk	pk
3.1.2. Ear protectors							pk	pk	pk	pk
3.1.3. Safety helmets							pk	pk	pk	pk
3.1.4. Respirators							pk	pk	pk	pk
3.1.5. Gloves							pk	pk	pk	pk
3.1.6. Safety boots							pk	pk	pk	pk
3.1.7. Seat belts							pk	pk	pk	pk
3.2. Cardiopulmonary resuscitation (CPR)							pk	pk	pk	pk
3.3. Employ safety principles when										
3.3.1. Getting on/off equipment							K	K	K	K
3.3.2. Starting/stopping engines							K	K	K	K
3.3.3. Placing equipment in motion							K	K	K	K
3.3.4. Operating in hazardous conditions							K	K	K	K
3.3.5. Servicing equipment							K	K	K	K
3.3.6. Loading equipment							K	K	K	K
3.3.7. Unloading equipment							K	K	K	K
3.3.8. Hauling equipment							K	K	K	K
3.3.9. Adjusting equipment							K	K	K	K
3.3.10. Loading material							K	K	K	K
3.3.11. Unloading material							K	K	K	K
3.3.12. Hauling material							K	K	K	K
3.3.13. Electrical hazards and precautions							K	K	K	K
3.4. Initial Federal Hazard Communication Training Program (FHCTP) **TR: DoD 6050.5; AFI 90-821**										
3.4.1. Traffic control							K	K	K	K
3.4.2. Worksite safety							K	K	K	K
3.4.3. Confined space health hazards							K	K	K	K
3.4.4. Pavement repair health hazards							K	K	K	K
4. AFS Specific Publications: **TR: TO 32-1-101, AFI 24-301**										
4.1. Use technical orders/ commercial manuals and/or AF Form 1800 to										
4.1.1. Perform operational checks							pk	pk	pk	pk
4.1.2. Perform maintenance							pk	pk	pk	pk
4.1.3. Locate part numbers in illustrated parts breakdown							K	K	K	K
4.1.4. Find vehicle/equipment capacities, capabilities, parameters							K	K	K	K

5. Communications **TR: AFIs 13-204 V3, 13-213, 33-106** **and** **AFVA 13-221**										
5.1. Use radios							pk	pk	pk	pk
5.2. Use hand signals							pk	pk	pk	pk
6. Project Planning/Management Fundamentals **TR: NTRP 4-04.2.5, TM 3-34.62, AFI** **32-1023, USAF Project Managers Guide for Design and Construction (AFCEC), UFC 3-740-05**										
6.1. Establish resource requirements										
6.1.1. Identify Equipment requirements								K	K	K
6.1.2. Identify Material requirements								K	K	K
6.1.3. Interpret construction drawings									K	K
6.1.4. Construction Stake Markings									K	K
6.2. Soil fundamentals: **TR: UFC 3-220-10N and 3-250-11; TM 3-34.64**										
6.2.1. Soil types									K	K
6.2.2. Soil construction properties									K	K
6.2.3. Soil stabilization techniques									K	K
6.3. Compute material requirements for										
6.3.1. Fill								K	pk	pk
6.3.2. Subgrade/Base course								K	pk	pk
6.3.3. Asphalt (hot and cold mix)								K	pk	pk
6.3.4. Concrete								K	pk	pk
7. Powered Equipment: **TR: TM 3-34.62, TO 34Y series and applicable owner's manuals**										
7.1. Perform operational checks							pk	pk	pk	pk
7.2. Perform operator maintenance							pk	pk	pk	pk
7.3. Use powered equipment										
7.3.1. Vibratory tamper							P	P	P	P
7.3.2. Pavement router (vertical/impact)							K	K	K	K
7.3.3. Crack saw							K	K	K	K
7.3.4. Portable concrete mixer							P	P	P	P
7.3.5. Power screed							K	P	P	P
7.3.6. Concrete saw							P	P	P	P
7.3.7. K-12 saw							P	P	P	P
7.3.8. Hot/cold joint sealing machine							P	P	P	P
7.3.9. Chain Saw							P	P	P	P
7.3.10. Saw Blades										
7.3.10.1. Asphalt							P	P	P	P
7.3.10.2. Concrete (cured/green)							P	P	P	P
7.3.10.3. Wet/dry							P	P	P	P
7.4. Air compressor										
7.4.1. Perform operational checks							pk	pk	pk	pk
7.4.2. Perform operator's maintenance							pk	pk	pk	pk
7.4.3. Operate							P	P	P	P

7.4.4. Use air compressor accessories										
7.4.4.1. Pneumatic drill									K	K
7.4.4.2. Pneumatic hammer with attachments							P	P	P	P
7.4.4.3. Pneumatic tamper								K	K	K
8. Earth Moving and Excavation Equipment: **TR: TM 3-34.62, STP 5- 6J12, TC 21-305-20, AFMAN 24- 306(I)**										
8.1. Dump trucks:										
8.1.1. Perform operational checks							pk	pk	pk	pk
8.1.2. Perform operator's maintenance							pk	pk	pk	pk
8.1.3. Operate										
8.1.3.1. Haul material							P	P	P	P
8.1.3.2. Dump material							P	P	P	P
8.1.3.3. Spread material							P	P	P	P
8.1.3.4. Tow equipment							P	P	P	P
8.1.3.5. Tailgate										
8.1.3.5.1. Removal							P	P	P	P
8.1.3.5.2. Installation							P	P	P	P
8.2. Wheel mounted front-end loaders										
8.2.1. Perform operational checks							pk	pk	pk	pk
8.2.2. Perform operator's maintenance							pk	pk	pk	pk
8.2.3. Operate										
8.2.3.1. Load material into dump truck							pk	pk	pk	pk
8.2.3.2. Level area								P	P	P
8.2.3.3. Spread materials								P	P	P
8.2.3.4. Stockpile materials							P	P	P	P
8.2.3.5. Backfill								P	P	P
8.2.3.6. Clamshell bucket								P	P	P
8.2.3.7. Excavate								P	P	P
8.2.4. Operate with forklift attachment										
8.2.4.1. Load/unload materials							P	P	P	P
8.2.4.2. Transport materials							P	P	P	P
8.2.5. Move material using										
8.2.5.1. Boom							pk	pk	pk	pk
8.2.5.2. Blade							pk	pk	pk	pk
8.2.6. Change attachments on loaders							P	P	P	P
8.3. Skid steer loader										
8.3.1. Perform operational checks								pk	pk	pk
8.3.2. Perform operator’s maintenance								pk	pk	pk

8.3.3. Operate using attachments such as										
8.3.3.1. Impactor								P	P	P
8.3.3.2. Bucket								P	P	P
8.3.3.3. Forklift								P	P	P
8.3.3.4. Earth auger								K	K	K
8.3.3.5. Backhoe								K	K	K
8.3.3.6. Sweeper								K	K	K
8.3.4. Perform functions such as										
8.3.4.1. Excavating								P	P	P
8.3.4.2. Backfilling								P	P	P
8.3.4.3. Leveling								P	P	P
8.3.4.4. Load materials								P	P	P
8.3.5. Change attachments on skid steer loader								P	P	P
8.4. Motorized Scrapers **TR: 36C22-2/3 Series**										
8.4.1. Perform operational checks								K	K	K
8.4.2. Perform operator's maintenance								K	K	K
8.4.3. Operate										
8.4.3.1. Excavate								K	K	K
8.4.3.2. Haul material								K	K	K
8.4.3.3. Dump material								K	K	K
8.4.3.4. Spread material								K	K	K
8.5. Graders										
8.5.1. Perform operational checks								pk	pk	pk
8.5.2. Perform operator's maintenance								pk	pk	pk
8.5.3. Perform grader operations such as										
8.5.3.1. Ditching								K	K	K
8.5.3.2. Sloping banks								K	K	K
8.5.3.3. Spreading materials in articulated mode								P	P	P
8.5.3.4. Leveling materials								P	P	P
8.5.3.5. Maintaining unpaved surfaces								P	P	P
8.5.3.6. Finish grading									P	P
8.5.3.7. Scarifying materials								P	P	P
8.5.3.8. Mixing materials								P	P	P
8.6. Crawler tractor										
8.6.1. Perform operational checks								pk	pk	pk
8.6.2. Perform operator's maintenance								pk	pk	pk

8.6.3. Perform crawler tractor operations such as										
8.6.3.1. Stockpiling								P	P	P
8.6.3.2. Leveling								P	P	P
8.6.3.3. Ditching								K	K	K
8.6.3.4. Ripping								K	K	K
8.6.3.5. Winching								K	K	K
8.6.3.6. Clearing								K	K	K
8.6.3.7. Bank sloping								K	K	K
8.6.3.8. Tow equipment								K	K	K
9. Material Handling Equipment										
9.1. Tractor trailer combinations: **TR: Title 49 CFR, Federal Motor Carrier Safety Administration, AFI 24- 301, TC 21-305-20 [AFMAN 24-306(I)], TC 21-305-100, STP 5- 62J12**										
9.1.1. Perform operational checks									pk	pk
9.1.2. Perform operator's maintenance									pk	pk
9.1.3. Operate										
9.1.3.1. Couple/uncouple trailer									P	P
9.1.3.2. Backing									P	P
9.1.3.3. Over the road									P	P
9.1.4. Transport and secure equipment/materials										
9.1.4.1. Load/unload construction equipment									pk	pk
9.1.4.2. Secure construction equipment using chains and binders									pk	pk
9.1.4.3. Load/unload construction material									pk	pk
9.1.4.4. Secure construction materials using cargo straps									pk	pk
9.1.4.5. Utilize cribbing									pk	pk
9.2. Forklift (FL): **TR: AFI 91-203, AFQTP (Forklift)**										
9.2.1. Perform operational checks							pk	pk	pk	pk
9.2.2. Perform operator's maintenance							pk	pk	pk	pk
9.2.3. Move material/ equipment using										
9.2.3.1. All Terrain (AT) FL							P	pk	pk	pk
9.2.3.2. Under 10K FL							P	pk	pk	pk
9.2.3.3. 10K and over FL							K	K	K	K
9.2.3.4. 463L capable							K	K	K	K

9.3. Hydraulic telescopic cranes **TR: FM 5-434, AFI 91-203, TOs 36C3/23, 36Y2**										
9.3.1. General										
9.3.1.1. Perform operational checks									pk	pk
9.3.1.2. Perform operator's maintenance									pk	pk
9.3.1.3. Operate crane carrier									pk	pk
9.3.1.4. Interpret load charts									pk	pk
9.3.1.5. Setup and maneuver									pk	pk
9.3.1.6. Hand signals								K	K	K
9.3.1.7. Safety requirements								K	K	K
9.3.2. Operate with										
9.3.2.1. Multi-parts-line Hook block								K	K	K
9.3.2.2. Boom extension (jib)								K	K	K
9.3.2.3. Single-line hook (headache ball)								K	K	K
9.3.2.4. Dielectric boom shield								K	K	K
9.3.2.5. Clamshell								K	K	K
9.3.3. Install and remove cables								K	K	K
10. Rigging and Lifting: **TR: FM 5-434 & 5-125, AFI 91-203**										
10.1. Identify the handling characteristics and inspection requirements of										
10.1.1. Chains									pk	pk
10.1.2. Wire ropes									pk	pk
10.1.3. Nylon slings									pk	pk
10.2. Rig loads to be lifted using										
10.2.1. Chains								P	P	P
10.2.2. Wire rope								P	P	P
10.2.3. Nylon slings								P	P	P
10.2.4. Spreader bars								P	P	P
10.2.5. Lifting beams								P	P	P
10.3. Move/manipulate loads using										
10.3.1. Taglines								P	P	P
11. Support Equipment										
11.1. Industrial tractors										
11.1.1. Perform operational checks								K	K	K
11.1.2. Perform operator's maintenance								K	K	K
11.1.3. Operate with attachments such as										
11.1.3.1. Dragbox								K	K	K
11.1.3.2. Earth auger								K	K	K
11.1.4. Tow equipment								K	K	K

11.2. Asphalt pavers										
11.2.1. Perform operational checks								K	K	K
11.2.2. Perform operator's maintenance								K	K	K
11.2.3. Plan paving operations								K	K	K
11.2.4. Operate self-propelled paver								K	K	K
11.3. Liquid Material Distributors **TR: TO 36C5 and 36C20 Series, STP 5- 62J12**										
11.3.1. Water distributor:										
11.3.1.1. Perform operational checks								P	P	P
11.3.1.2. Perform operator's maintenance								P	P	P
11.3.1.3. Perform liquid material filling operations										
11.3.1.3.1. Gravity feed								P	P	P
11.3.1.3.2. Pressure spray								P	P	P
11.3.1.3.3. Fill from hydrant								P	P	P
11.3.1.3.4. Fill from stand								P	P	P
11.3.1.3.5. Suction fill								K	K	K
11.4. Sweepers: **TR: TO 36C25 Series**										
11.4.1. Towed sweepers										
11.4.1.1. Perform operational checks								P	P	P
11.4.1.2. Perform operator's maintenance								P	P	P
11.4.1.3. Operate								P	P	P
11.4.2. Front-mounted sweepers										
11.4.2.1. Perform operational checks								P	P	P
11.4.2.2. Perform operator's maintenance								P	P	P
11.4.2.3. Operate								P	P	P
11.4.3. Multi-purpose sweepers										
11.4.3.1. Perform operational checks							pk	pk	pk	pk
11.4.3.2. Perform operator's maintenance							pk	pk	pk	pk
11.4.3.3. Perform sweeper maintenance										
11.4.3.3.1. Change brooms							P	P	P	P
11.4.3.3.2. Change Wearing Surfaces										
11.4.3.3.2.1. Skid plates							P	P	P	P
11.4.3.3.2.2. Curtains							P	P	P	P
11.4.3.3.3. Adjust hood							P	P	P	P
11.4.3.3.4. Adjust broom							P	P	P	P
11.4.3.3.5. Clean sweeper							P	P	P	P

11.4.3.4. Airfield Operations **TR: AFI 13-213**										
11.4.3.4.2. Street							P	P	P	P
11.4.3.5. Operate suction hose attachment							P	P	P	P
11.5. Compaction Equipment TR: TO 36C2O Series										
11.5.1. Perform operational checks								pk	pk	pk
11.5.2. Perform operator's maintenance								pk	pk	pk
11.5.3. Compact earthen materials using										
11.5.3.1. Pneumatic tired roller								P	P	P
11.5.3.2. Sheepsfoot roller								K	K	K
11.5.3.3. Steel wheel vibratory roller								P	P	P
12. Excavating/Trenching Machines **TR: TO 36C6, 36C12 & 36C37 Series, FM 5- 434, STP 5-62J12, AFI 91-203**										
12.1. Shoring applications								K	K	K
12.2. Backhoe										
12.2.1. Perform operational checks								pk	pk	pk
12.2.2. Perform operator's maintenance								pk	pk	pk
12.2.3. Perform backhoe operations such as										
12.2.3.1. Excavate with front bucket									P	P
12.2.3.2. Excavate with rear bucket								P	P	P
12.2.3.3. Partial depth removal								P	P	P
12.2.3.4. Load with front bucket								P	P	P
12.2.3.5. Load with rear bucket								P	P	P
12.2.3.6. Backfill with front bucket								P	P	P
12.2.3.7. Backfill with rear bucket									P	P
12.2.3.8. Break material with Impactor							pk	pk	pk	pk
12.3. Hydraulic excavator										
12.3.1. Perform operational checks								pk	pk	pk
12.3.2. Perform operator's maintenance								pk	pk	pk
12.3.3. Perform excavator operations such as										
12.3.3.1. Excavate with bucket								pk	pk	pk
12.3.3.2. Load with bucket								pk	pk	pk
12.3.3.3. Backfill with bucket/dozer blade								pk	pk	pk
12.3.3.4. Compact materials								pk	pk	pk
12.3.3.5. Break materials with Impactor								P	P	P
12.3.3.6. Lift/position loads								P	P	P
12.3.4. Change excavator attachments such as										
12.3.4.1. Bucket								P	P	P
12.3.4.2. Impactor								P	P	P
12.3.4.3. Compactor								P	P	P

12.4. Trenching machines										
12.4.1. Perform operational checks								P	P	P
12.4.2. Perform operator's maintenance								P	P	P
12.4.3. Perform operations such as										
12.4.3.1. Trenching								P	P	P
12.4.3.2. Backfilling								P	P	P
13. Snow and Ice Control: **TR: AFI 32-1002 & 13-204 V3**										
13.1. Snow and ice control procedures										
13.1.1. Snow and Ice Control Plan								K	K	K
13.1.2. Airfield paved surfaces								K	K	K
13.1.3. Non-airfield surfaces								K	K	K
13.2. Remove snow using										
13.2.1. Snow blowers								P	P	P
13.2.2. Snowplows								P	P	P
13.2.3. Snow sweepers								P	P	P
13.2.4. Multi-purpose snow removal unit								P	P	P
13.3. Perform ice control operations using										
13.3.1. Mechanical equipment								K	K	K
13.3.2. Chemicals								K	K	K
13.3.3. Abrasives								K	K	K
13.4. Runway Ice Detection System								K	K	K
13.5. Snow & Ice Control System								K	K	K
13.6. Snow blowers										
13.6.1. Perform operational checks								P	P	P
13.6.2. Perform operator's maintenance								P	P	P
13.6.3. Operate								P	P	P
13.7. Snowplows										
13.7.1. Perform operational checks								P	P	P
13.7.2. Perform operator's maintenance								P	P	P
13.7.3. Operate								P	P	P
13.8. Snow sweepers										
13.8.1. Perform operational checks								P	P	P
13.8.2. Perform operator's maintenance								P	P	P
13.8.3. Operate								P	P	P
13.9. Multi-purpose snow removal unit										
13.9.1. Perform operational checks								P	P	P
13.9.2. Perform operator's maintenance								P	P	P
13.9.3. Operate								P	P	P
13.9.4. Change attachments								P	P	P
13.10. Environmental concerns										
13.10.1. Chemicals								K	K	K
13.10.2. Abrasives								K	K	K

14. Construct Paved Surfaces: TR: UFCs 3-250-01/03/04, 3-270-08, FMs 5-428, 5-436 ETL 11-26, Asphalt Institute, The Asphalt Handbook, Portland Cement Association, Design and Control of Concrete Mixtures										
14.1. Layout the area using										
14.1.1. Line level							P	P	P	P
14.1.2. Laser level							K	K	K	K
14.1.3. Carpenter hand level							P	P	P	P
14.2. Prepare area as indicated by construction stakes										
14.2.1. Clear the area using hand tools							P	P	P	P
14.2.2. Clear the area using construction equipment							P	P	P	P
14.2.3. Prepare subgrade							P	P	P	P
14.2.3.1. Stabilize using										
14.2.3.1.1. Mechanical							K	K	K	K
14.2.3.1.2. Chemical							K	K	K	K
14.2.3.2. Compact							P	P	P	P
14.2.4. Prepare base course										
14.2.4.1. Place materials							P	P	P	P
14.2.4.2. Compact materials							P	P	P	P
14.3. Rigid pavement/Portland Cement Concrete (PCC)										
14.3.1. Types of concrete materials								K	K	K
14.3.2. Standard concrete mix							P	P	P	P
14.3.3. Construct concrete slab										
14.3.3.1. Install forms							P	P	P	P
14.3.3.2. Lubricate forms before use							P	P	P	P
14.3.3.3. Install reinforcing materials										
14.3.3.3.1. Wire mesh								P	P	P
14.3.3.3.2. Rebar								P	P	P
14.3.3.4. Install load transfer devices:										
14.3.3.4.1. Keyways								K	K	K
14.3.3.4.2. Dowels								K	K	K
14.3.3.5. Place concrete							P	P	P	P
14.3.3.6. Fabricate joints								P	P	P
14.3.3.7. Finish concrete										
14.3.3.7.1. Trowel								P	P	P
14.3.3.7.2. Non-Skid								P	P	P
14.3.3.8. Cure concrete								P	P	P
14.3.3.9. Remove/clean/oil forms							P	P	P	P
14.3.3.10. Landscape site							P	P	P	P

14.3.4. Repair defective rigid pavement **TR: UFC 3-250-06, 3-270-04**										
14.3.4.1. Perform full depth pavement repair										
14.3.4.1.1. Identify defective pavement							P	P	P	P
14.3.4.1.2. Cut pavement using concrete saw							P	P	P	P
14.3.4.1.3. Break pavement jackhammer							P	P	P	P
14.3.4.1.4. Remove pavement							P	P	P	P
14.3.4.2. Prepare the area										
14.3.4.2.1. Remove/replace base course							P	P	P	P
14.3.4.2.2. Remove/replace subgrade							P	P	P	P
14.3.4.2.3. Replace load transfer devices								P	P	P
14.3.4.2.4. Replace reinforcement material								P	P	P
14.3.4.3. Replace concrete										
14.3.4.3.1. Mix concrete using										
14.3.4.3.1.1. Hand tools							P	P	P	P
14.3.4.3.1.2. Portable concrete mixer							P	P	P	P
14.3.4.3.2. Expansion material							P	P	P	P
14.3.4.3.3. Place concrete							P	P	P	P
14.3.4.3.4. Finish concrete								P	P	P
14.3.4.3.5. Cure concrete								P	P	P
14.3.5. Perform partial depth/spall repair **TR: UFC 3-270-03**										
14.3.5.1. Identify defective pavement							P	P	P	P
14.3.5.2. Cut pavement using concrete saw							P	P	P	P
14.3.5.3. Break pavement using jack hammer							P	P	P	P
14.3.5.4. Remove pavement							P	P	P	P
14.3.5.5. Mix/place patch materials							P	P	P	P
14.3.6. Repair defective sealing material: **TR: UFC 3-250-08FA**										
14.3.6.1. Prepare joints/cracks for sealing										
14.3.6.1.1. Remove existing sealant using										
14.3.6.1.1.1. Vertical router								P	P	P
14.3.6.1.1.2. Impact router								P	P	P
14.3.6.1.1.3. Crack saw								P	P	P
14.3.6.2. Saw or reface the joint using concrete saw								P	P	P

14.3.6.3. Clean joints/cracks with										
14.3.6.3.1. High pressure water							P	P	P	P
14.3.6.3.2. High pressure air							P	P	P	P
14.3.6.4. Reseal joints/cracks										
14.3.6.4.1. Install backer materials							P	P	P	P
14.3.6.4.2. Apply cold sealants							P	P	P	P
14.3.6.4.3. Apply Hot sealants							P	P	P	P
14.4. Flexible pavements/Asphalt Cement Concrete										
14.4.1. Construct with bituminous materials (new construction)										
14.4.1.1. Types								K	K	K
14.4.1.2. Uses								K	K	K
14.4.1.3. Environmental concerns								P	P	P
14.4.1.4. Apply prime coat								P	P	P
14.4.1.5. Determine asphalt temperatures								P	P	P
14.4.1.6. Place asphalt								P	P	P
14.4.1.7. Asphalt concrete lifts										
14.4.1.7.1. Binder course								K	K	K
14.4.1.7.2. Wear course								K	K	K
14.4.1.8. Compact asphalt							P	P	P	P
14.4.2. Repair defective pavement **TR: UFC 3-270-01:**										
14.4.2.1. Identify the area							P	P	P	P
14.4.2.2. Cut area							P	P	P	P
14.4.2.3. Remove pavement							P	P	P	P
14.4.2.4. Prepare subgrade							P	P	P	P
14.4.2.5. Prepare base							P	P	P	P
14.4.2.6. Apply prime/tack coat							P	P	P	P
14.4.2.7. Place asphalt							P	P	P	P
14.4.2.8. Compact asphalt							P	P	P	P
14.4.3. Repair cracks in flexible pavements: **TR: UFC 3-250-08FA, 3- 270-02**										
14.4.3.1. Prepare cracks using										
14.4.3.1.1. Vertical router							P	P	P	P
14.4.3.1.2. Impact router							P	P	P	P
14.4.3.1.3. Crack saw							P	P	P	P
14.4.3.1.4. Clean area using high pressure air							P	P	P	P
14.4.3.2. Seal cracks using										
14.4.3.2.1. Hot applied sealant							P	P	P	P
14.4.3.2.2. Cold applied sealant							P	P	P	P

14.4.3.2.3. Apply surface treatments										
14.4.3.2.3.1. Fog seal								K	K	K
14.4.3.2.3.2. Slurry seal								K	K	K
14.4.3.2.3.3. Chip seal								K	K	K
15. Drainage Systems: **TR: ETL 04-6, UFC 3-260-02 & 3-230- 17FA**										
15.1. Surface drainage										
15.1.1. Construct								P	P	P
15.1.2. Inspect								P	P	P
15.1.3. Repair								P	P	P
15.2. Sub-surface drainage										
15.2.1. Install								P	P	P
15.2.2. Inspect (operational)								P	P	P
15.2.3. Repair								P	P	P
16. Fences: **TR: AFMAN 32-1084, AFI 31-101**										
16.1. Install:										
16.1.1. Chain link							pk	pk	pk	pk
16.1.2. Barbed wire							pk	pk	pk	pk
16.1.3. Snow fence							pk	pk	pk	pk
16.2. Inspect:										
16.2.1. Chain link							pk	pk	pk	pk
16.2.2. Barbed wire							pk	pk	pk	pk
16.3. Repair:										
16.3.1. Chain link							P	P	P	P
16.3.2. Barbed wire							P	P	P	P

Attachment 3 – Air Force Qualification Training Package (AFQTP) Tracker

A3. AFQTP Documentation Record

A3.1. To ensure each Pavements and Construction Equipment Operator Specialist is trained to the correct standard an AFQTP has been developed for each task group identified in their STS. These AFQTPs are to be used by the employee, trainer and certifier in their on-the-job training program for qualification training and developmental training.

A3.2. These AFQTPs ensure all aspects of the task are covered sufficiently and provide additional task knowledge, in preparation for hands-on training. AFQTPs summarize procedures on a task performance checklist for use by trainers, certifiers, and employees.

A3.2.1. The UTM or supervisor can download paper based AFQTP's. Paper-based AFQTP's are found on CE DASH under documents in the AFQTP folder.

A3.2.2. In addition to the paper-based AFQTPs there are web-based courses or assessments developed for certain tasks that are available on myLearning under AFCEC in the Home Station topic area.

A3.3. Documentation. Before a core or diamond task can be signed off in the JQS section of the individual automated training record, the task must be signed off in the QTP section first.

A3.4. AFQTP Tracker

Task Number	Tasks, Knowledge, and Technical References	Tasks		Certification of AFQTPs			
		A/J	C/S	Tng Start	Tng Complete	Trainee Initials	Trainer Initials
1.0.	**Civil Engineer Core Concepts**						
1.1.	Complete CE 5-Level Core Concepts Course	X					
1.2.	Complete CE 7-Level Core Concepts Course	X					
1.4.	**Asset Management**						
1.4.1.	Complete AFIT WMGT 301 Intro to Asset Management	X					
4.0.	**Communications**						
4.1.	Demonstrate proper use of radios	X					
4.2.	Demonstrate proper use of hand signals	X					
5.0.	**Project Planning / Management Fundamentals**						
5.1.	**Resource requirements establishment**						
5.1.3.	Demonstrate proper use of construction drawings		X				
5.2.	**Soil Fundamentals**						
5.2.1.	Soil types		X				
5.2.2.	Soil construction properties		X				
5.2.3.	Soil stabilization techniques		X				
5.3.	**Compute material requirements for**						
5.3.1.	Fill	X					
5.3.2.	Base course	X					
5.3.3.	Asphalt (hot and cold mix)	X					
5.3.4.	Concrete	X					
5.4.	**Project management web-based courses**						
5.4.1.	Complete AFIT WMGT 322 Intro to Project Management		X				
6.0.	**Powered Equipment / Tools/ Units of Measure**						
6.4.4.	**Air Compressor**						
6.4.4.1.	Operational checks	X					
6.4.4.2.	Operator maintenance	X					
6.4.4.3.	Operate	X					
6.4.5.	**Demonstrate proper use of air compressor accessories**						
6.4.5.2.	Pneumatic hammer with attachments	X					
7.0.	**Earth-Moving and Excavation Equipment**						
7.2.	**Dump Truck**						
7.2.1.	Perform operational checks	X					
7.2.2.	Perform operator maintenance	X					
7.2.3.	**Operate to**						
7.2.3.1.	Haul Material	X					

Task Number	Tasks, Knowledge, and Technical References	Core/Diamond Tasks		Certification of AFQTPs			
		A/J	C/S	Tng Start	Tng Complete	Trainee Initials	Trainer Initials
7.2.3.2.	Dump material	X					
7.2.3.3.	Spread material	X					
7.3.	**Wheel-mounted front-end loader**						
7.3.1.	Perform operational checks	X					
7.3.2.	Perform operator maintenance	X					
7.3.3.	**Operate to**						
7.3.3.1.	Load material into dump truck	X					
7.3.3.2.	Level area (spread)	X					
7.3.3.3.	Stockpile material	X					
7.3.3.4	Backfill	X					
7.3.3.5.	Utilize clamshell bucket	X					
7.3.3.6.	Excavate	X					
7.3.4.	**Operate with forklift attachment to:**						
7.3.4.1.	Load/unload material	X					
7.3.4.2.	Transport material	X					
7.3.6.	Change loader attachments	X					
7.4.	**Compact Track Loader (CTL)**						
7.4.1.	Perform operational checks	X					
7.4.2.	Perform operator maintenance	X					
7.4.3.	**Operate using attachments:**						
7.4.3.1.	**Bucket**						
7.4.3.1.1.	To excavate	X					
7.4.3.1.2.	To backfill	X					
7.4.3.1.3.	To level	X					
7.4.3.1.4.	To load material	X					
7.4.4.	Change CTL attachments	X					
7.6.	**Graders**						
7.6.1.	Perform operational checks	X					
7.6.2.	Perform operator maintenance	X					
7.6.3.	**Perform grader operations such as:**						
7.6.3.3.	Spreading material	X					
7.6.3.4.	Leveling material	X					
7.6.3.5.	Maintaining unpaved surfaces	X					
7.6.3.6.	Finish grading		X				
7.7.	**Crawler tractor**						
7.7.1.	Perform operational checks		X				
7.7.2.	Perform operator maintenance		X				
7.7.3.	**Perform crawler tractor operations such as:**						
7.7.3.1.	Stockpiling		X				
7.7.3.2.	Leveling		X				
8.0.	**Material-Handling Equipment**						
8.1.4.	**Equipment/materials securement and transportation**						
8.1.4.1.	Load construction equipment	X					
8.1.4.2.	Unload construction equipment	X					

Task Number	Tasks, Knowledge, and Technical References	Core/Diamond Tasks		Certification of AFQTPs			
		A/J	C/S	Tng Start	Tng Complete	Trainee Initials	Trainer Initials
8.1.4.3.	Secure construction equipment using chains and binders	X					
8.1.4.4.	Load construction material	X					
8.1.4.5.	Unload construction material	X					
8.1.4.6.	Secure construction material using cargo straps	X					
9.0.	**Hydraulic Mobile Crane**						
9.1.	**Rigging and lifting**						
9.1.1.	**Principles of rigging**						
9.1.1.1.	Perform load estimation	X					
9.1.1.2.	Determine center of gravity	X					
9.1.1.3.	Calculate sling requirements (weight, length, angle)	X					
9.1.2.	**Handling characteristics and inspection requirements of:**						
9.1.2.1.	Chains	X					
9.1.2.2.	Wire ropes	X					
9.1.2.3.	Nylon slings	X					
9.1.2.4	Shackles	X					
9.1.2.5.	Hooks	X					
9.1.3.	**Rig loads to be lifted using**						
9.1.3.1.	Chains	X					
9.1.3.2.	Wire ropes	X					
9.1.3.3.	Nylon slings	X					
9.1.3.4.	Spreader bars	X					
9.1.3.6.	Taglines	X					
9.2.	**Crane operations**						
9.2.1.	**General**						
9.2.1.4.	Interpret load charts	X					
9.2.1.7.	Perform crane hand signals	X					
9.2.1.8.	Perform safety requirements	X					
10.0.	**Support Equipment**						
10.3.	**Liquid-material distributors**						
10.3.1.	**Water distributor**						
10.3.1.1.	Perform operational checks	X					
10.3.1.2.	Perform operator maintenance	X					
10.3.1.3.	**Perform operations such as:**						
10.3.1.3.1.	Gravity feed	X					
10.3.1.3.2.	Pressure spray	X					
10.3.1.3.3.	Fill from hydrant	X					
10.3.1.3.4.	Fill from stand	X					
10.4.	**Sweepers**						

Task Number	Tasks, Knowledge, and Technical References	Core/Diamond Tasks		Certification of AFQTPs			
		A/J	C/S	Tng Start	Tng Complete	Trainee Initials	Trainer Initials
10.4.1.	**Multi-purpose sweepers**						
10.4.1.1.	Perform operational checks	X					
10.4.1.2.	**Operator maintenance**						
10.4.1.2.1.	Adjust hood	X					
10.4.1.2.2.	Adjust broom	X					
10.4.1.2.3.	Clean sweeper	X					
10.4.1.2.4.	**Change wearing surfaces**						
10.4.1.2.4.1.	Brooms	X					
10.4.1.2.4.2.	Skid plates	X					
10.4.1.2.4.3.	Curtains	X					
10.4.2.	**Operate on:**						
10.4.2.1.	Airfield	X					
10.4.2.2.	Street	X					
10.4.3.	Operate suction hose attachment	X					
10.5.	**Compaction equipment**						
10.5.1.	Perform operational checks	X					
10.5.2.	Perform operator maintenance	X					
10.5.3.	**Compact earthen materials using:**						
10.5.3.3.	Steel wheel vibratory roller	X					
11.0.	**Excavating/Trenching Machines**						
11.1.	**Backhoe**						
11.1.1.	Perform operational checks	X					
11.1.2.	Perform operator maintenance	X					
11.1.3.	**Perform backhoe operations such as:**						
11.1.3.2.	Excavating with rear bucket	X					
11.1.3.4.	Loading with rear bucket	X					
11.1.3.5.	Backfilling with front bucket	X					
11.1.3.6.	Backfilling with rear bucket	X					
11.2.	**Hydraulic excavator**						
11.2.1.	Perform operational checks	X					
11.2.2.	Perform operator maintenance	X					
11.2.3.	**Perform excavator operations such as:**						
11.2.3.1.	Excavating with bucket	X					
11.2.3.2.	Loading with bucket	X					
11.2.3.3.	Backfilling with bucket	X					
11.2.3.5.	Breaking materials with impactor	X					
11.2.4.	Change excavator attachments	X					
13.0.	**Paved Surfaces Construction**						
13.1.	**Lay out area using:**						
13.1.1.	Line level	X					
13.1.3.	Carpenter hand level	X					

Task Number	Tasks, Knowledge, and Technical References	Core/Diamond Tasks		Certification of AFQTPs			
		A/J	C/S	Tng Start	Tng Complete	Trainee Initials	Trainer Initials
13.2.	**Area preparation**						
13.2.1.	Clear the area using hand tools	X					
13.2.2.	Clear the area using construction equipment	X					
13.2.3.	Stabilize subgrade using chemical means		X				
13.2.4.	Stabilize subgrade using mechanical means	X					
13.2.5.	**Base course preparation**						
13.2.5.1.	Place materials	X					
13.2.5.2.	Compact materials	X					
13.2.6.	**Concrete slab construction**						
13.2.6.1.	Install forms	X					
13.3.	**Rigid pavement/Portland Cement Concrete (PCC)**						
13.3.3.	Perform concrete test (slump test)		X				
13.3.4.	Place concrete	X					
13.3.5.	**Finish concrete:**						
13.3.5.1.	Using power screed	X					
13.3.5.2.	Using float/trowel	X					
13.3.5.3.	By edging	X					
13.3.5.4.	By fabricating joints	X					
13.3.5.5.	By creating a non-skid surface	X					
13.3.5.6.	By curing concrete	X					
13.3.5.10.	By sealing joints	X					
13.4.	**Defective rigid pavement repair**						
13.4.1.	**Rigid pavement inspection**						
13.4.1.1.	Identify pavement defects	X					
13.4.1.2.	Pavement Condition Index (PCI)		X				
13.4.1.3.	**Full-depth pavement repair**						
13.4.1.3.2.	Cut pavement using concrete saw	X					
13.4.1.3.7.	Break pavement using air compressor with jack-hammer	X					
13.4.1.3.8.	Remove pavement	X					
13.4.3.	**Defective sealing material repair**						
13.4.3.1.	**Prepare joints/cracks for sealing**						
13.4.3.1.2.	Remove existing sealant using powered equipment	X					
13.4.3.3.	**Joints/cracks resealing**						
13.4.3.3.1.	Install backer material		X				
13.4.3.3.2.	Apply cold sealant		X				
13.4.3.3.3.	Apply hot sealant		X				
13.5.	**Flexible pavements/Asphalt Cement Concrete (ACC)**						
13.5.1.	**Construction with bituminous materials (new construction)**						
13.5.1.6.	Determine asphalt temperatures	X					
13.5.1.7.	Place asphalt	X					
13.5.1.8.	**Asphalt concrete lifts**						

Appendix A – Acronyms

Acronym	Definition
AETC	Air Education and Training Command
AETCI	Air Education and Training Command Instruction
AF	Air Force
AFCCFM	Air Force Civilian Career Field Manager
AFCDA	Air Force Career Development Academy
AFCEC	Air Force Civil Engineer Center
AFCEC/COF	Air Force Civil Engineer Center Force Development Division
AFIT	Air Force Institute of Technology
AFQTP	Air Force Qualification Training Package
AFSC	Air Force Specialty Code
ALS	Airman Leadership School
CBB	Career Building Block
CBT	Computer-Based Training
CDC	Career Development Course
CDL	Commercial Driver's License
CE	Civil Engineer
CFETP	Career Field Education and Training Plan
CFM	Career Field Management
COTS	Commercial Off The Shelf
DAFI	Department of Air Force Instruction
DAFMAN	Department of Air Force Manual
DL	Distance Learning
DoD	Department of Defense
EPME	Enlisted Professional Military Education
FSTS	Family Series Training Standard
FWS	Federal Wage System
HAF	Headquarters Air Force
HQ	Headquarters
IAW	In Accordance With
JQS	Job Qualification Standard
MTT	Mobile Training Team
NCOA	Noncommissioned Officer Academy
OJT	On-the-Job Training
PCs	Progress Checks
PME	Professional Military Education
SCPD	Standard Core Personnel Document
SEI	Special Experience Identifier
SNCOA	Senior Noncommissioned Officer Academy
SOS	Squadron Officers School
STS	Specialty Training Standard
TOC	Table of Contents
TR	Technical References

Acronym	Definition
USAF	United States Air Force
UETM	Unit Education Training Manager
UTM	Unit Training Manager
WBT	Web-Based Training

Appendix B - Master List of Links

Content	Location	Web Address
AETCI 36 - 2651, *Basic Military and Technical Training*	Part II, B4	https://static.e-publishing.af.mil/production/1/aetc/publication/aetci36-2651/aetci36-2651.pdf
AFIT Course List	Appendix C	https://www.afit.edu/ce/
Air Force Portal	Part I, B3 Part I, B3.1	https://www.my.af.mil/
Air University URL	Part I, B3.2	https://www.airuniversity.af.edu/GCPME/
CE Career Field Team SharePoint	Appendix C	https://usaf.dps.mil/sites/10016/Federal Wage System/Forms/AllItems.aspx?viewpath=%2Fsites%2F10016%2FFederal%20Wage%20System%2FForms%2FAllItems%2Easpx
CE DASH	Attachment 3, A3.2.1	https://usaf.dps.mil/teams/10758/CEtraining/SitePages/Home.aspx
CE Training Email	Part I, A4	afcec.ce.training@us.af.mil
Classification Standards	Appendix C	https://www.opm.gov/policy-data-oversight/classification-qualifications/classifying-federal-wage-system-positions/#url=5300
DAFI 36-2670, *Total Force Development*	Part I, B4 Part I, B4.1 Part II, A2 Part II, A2.6	https://static.e-publishing.af.mil/production/1/af_a1/publication/dafi36-2670/dafi36-2670.pdf
DAFMAN 32-2670, *Civil Engineer Federal Wage System Force Development*	Part II, A2 Part II, A2.6	https://static.e-publishing.af.mil/production/1/af_a4/publication/dafman32-2670/dafman32-2670_.pdf
DoD Civilian COOL	Appendix C	http://dod-civ-cool-review.s3.amazonaws.com/index.htm
Education and Training Course Announcements	Part II, D1 Appendix C	https://usaf.dps.mil/teams/app10-etca/SitePages/home.aspx?isdlg=1&IsDlg=1
Enlisted Professional Military Education (EPME)	Appendix C	https://www.airuniversity.af.edu/
Federal Building Competency Model	Attachment 2, A.2.2	https://www.gsa.gov/system/files/FBPTA%20Competencies%20Model%20update%20DEC%202024.xlsx
Job Series CBBs	Part I, B5.1.1	https://usaf.dps.mil/sites/10016/Career%20Building%20Blocks/Forms/AllItems.aspx
Civilian Force Development Homepage	Part II, D5 Appendix C	https://myfss.us.af.mil/USAFCommunity/s/knowledge-detail?xid=13085
myLearning	Part II, A2.6 Part II, C1.2.1 Part II, C1.2.2 Part II, C2.1 Attachment 3, A3.2.2 Appendix C	https://lms-jets.cce.af.mil/moodle/
Red Vector	Appendix C	https://afcec.redvector.com/lpe/course/search/b2b
FWS Series Qualification Standard (WGSQS)	Appendix C	https://www.opm.gov/policy-data-oversight/classification-qualifications/federal-wage-system-qualifications/%23url=List-of-Approved-Job-Elements

Appendix C - Resources

Resource/Term	Definition
Air Force Civil Engineer Center (AFCEC)	The focal point for all Civil Engineer training development.
Air Force Institute of Technology (AFIT)	Provides vital, relevant, and connected education that enables Airmen to be ready engineers and great leaders who know how to build sustainable installations to last while leading the change for the Civil Engineer career field. Review the Course list.
Air Force Qualification Training Package (AFQTP)	An instructional package designed for use as a training resource to qualify, or aid qualification, in a duty position or program, or on a piece of equipment. AFQTPs identify the AF's standardized method for performing the task. The AFQTP may be printed (paper-based), computer-based, in other audiovisual media formats, or all three.
Career Development Course (CDC)	A self-paced correspondence course published to provide the information necessary to satisfy the career knowledge component of OJT. These courses are developed from references identified in the CFETP. CDCs will contain information on basic principles, techniques, and procedures common to a military AFSC or civilian job series. They do not contain information on specific equipment or tasks unless best illustrating a procedure or technique having utility to the entire career field.
Career Field Education and Training Plan (CFETP)	A comprehensive, multipurpose document encapsulating the entire spectrum of education and training for various FWS series. It outlines a logical growth plan that includes training resources and is designed to make job series training identifiable, to eliminate duplication, and to ensure this training is budget defensible.
Career Field Manager (CFM)	An individual on the Air Staff charged with the responsibility for overseeing all training and career field management aspects of multiple AF job series in a functional area.
CE Career Field Team SharePoint	Contains information on leadership courses and civilian tuition assistance for degree programs. See: Civil Engineer Career Field Team SharePoint.
Civilian Force Development Homepage	Contains descriptions and requirements for several civilian training opportunities such as degree programs and civilian leadership courses. Click here to view the homepage.
Classification Standard	Job grading standards provide information used in determining the occupational series and title of jobs performing trades, craft, and labor work in the Federal Government. They also provide grading criteria for positions classified under the Federal Wage System (FWS). Click here to view Classification Standards.
Commercial Off the Shelf (COTS)	Commercially procured training products or in-resident vendor training.
Computer-Based Training (CBT)	A self-paced stand-alone computer product used to deliver interactive subject and task knowledge.

Resource/Term	Definition
Core Tasks	Mandatory tasks which the CFM has identified as a minimum qualification requirement within a job series or duty position. These tasks are derived from the Standard Core Personnel Document (SCPD) for each job series.
Distance Learning (DL)	Includes Video Tele-seminar (VTS), Video Tele-training (VTT), and CBT. Formal courses that a training wing or a contractor develops for export to a field location (in place of resident training) for trainees to complete without the on-site support of the formal school instructor. For instance, courses are offered by AFIT, Air University, and Training Detachments.
DoD Civilian COOL	On-line source for civilian credentialing opportunities.
Duty Position Tasks	Tasks identified by the work center supervisor as critical and common training tasks needed for the duty position and mission accomplishment.
Education & Training Course Announcements (ETCA)	Web platform, which contains descriptions, requirements, and reporting procedures for in resident Air Force courses.
Enlisted Professional Military Education (EPME)	EPME provides a continuum of learning through progressive courses concentrated on developing Military and Civilian Airmen. EPME plays a vital role in preparing Civilian Airmen for increased supervision, leadership, and management challenges. The three levels of AF EPME are Airman Leadership School, Noncommissioned Officer Academy and Air Force Senior Noncommissioned Officer Academy. All levels of EPME are available to FWS civilians in accordance with DAFI 36-2685 and DAFI 36-2687. Click here for more Information about EPME.
Family Series Training Standard (FSTS)	Describes skills and knowledge that FWS Employees in a particular job series need on the job and for future career development opportunities. It further serves as the overall training requirements for a Wage Series taught in the resident and nonresident courses as well as an on-the-job training program
Functional Advisory Council (FAC)/Federal Wage System Panel	The Federal Wage System Panel is one of the four panels that make up the Civil Engineer FAC. The Federal Wage System Panel charter is to work issues, develop policy, and provide recommendations to the FAC on matters related to civilian FWS requirements. The Federal Wage System Panel works through the FAC, in service to the CE Total Force community.
myLearning	Anytime, anywhere learning within DoD consisting of instructional modules comprised of sharable content objectives in an Internet/Intranet environment. Click here for mylearning.
On-the-Job Training (OJT)	Hands-on, over-the-shoulder training conducted to certify personnel in job qualification (duty position certification) training.
Proficiency Training	Additional training, either in-residence, advanced/supplemental training courses, or on-the-job training provided to personnel to increase their skills and knowledge beyond the minimum.

Resource/Term	Definition
Red Vector	Commercial web-based training available free of charge to CE employees. Courses are effective for certification renewal and earn Continuing Educations Units (CEUs). Click here for Red Vector site.
Regional Training Site (RTS)	Total Force training centers managed by the Air National Guard. These sites offer training on specialized military equipment and are available to civilians who require training for local mission needs. Training can be coordinated through the FWS Force Development Team at AFCEC/COF.
Resource Constraints	Resource deficiencies, such as money, facilities, time, manpower, or equipment that precludes desired training from being delivered.
Total Force	All collective AF components (Active Duty, Reserve, Guard, and Civilian elements) of the USAF.
Vendor Training	Training provided to the Air Force by a third party. Typically, a private vendor in the private sector not affiliated with the Department of Defense. Vendor training can be in the form of Web-Based Training, Computer-Based Training, or in resident training at a temporary duty location or even hosted on an Air Force installation.
Federal Wage System Series Qualification Standard (WGSQS)	A comprehensive task list that describes a particular series or duty position. Used by supervisors to document task qualifications. The tasks on the AFWGSQS are common to all persons serving in the described duty position. Click here for qualifications.
FWS Series Training	A mix of formal training (technical school) and informal training (on-the-job) to develop, maintain and enhance FWS series specific technical skills.
Web-Based Training	A form of Distance Learning. The term Web-Based just means the training is online and requires access to the internet in addition to the actual course content.